SECTION 26 32 13

DIESEL GENERATOR SETS

TECHNICAL SPECIFICATION

Project	MERIDIAN-1 Data Centre, Hall A
Project code	MER-1-2026
Issue	Issued for Construction — Revision C
Date	17 February 2026

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PART 1 — GENERAL

1.1 SUMMARY

A. Provide a complete, coordinated system including engineering, manufacture, delivery, installation support, testing, commissioning, training, and closeout.

B. Coordinate clearances, structural loads, electrical supplies, controls, fire strategy, access, and phased energisation.

1.2 REFERENCES

A. Comply with the following referenced standards.

1. ISO 8528-1:2018
2. ISO 3046-1:2002
3. CPCB IV+

1.3 DEFINITIONS

A. “Approved” means accepted for the stated stage without relieving the Contractor of responsibility.

B. “Provide” means design, supply, install, connect, test, commission, and document.

PART 1 — GENERAL

1.4 ACTION SUBMITTALS

A. Submit project-specific product data, selected ratings, dimensions, weights, clearances, and connection requirements.

B. Submit a clause-by-clause compliance statement identifying every exception.

C. Submit certified calculations and performance data at scheduled conditions.

1.5 INFORMATIONAL SUBMITTALS

A. Submit type-test certificates traceable to the offered construction.

B. Submit quality certificates, calibration records, and proposed factory inspection and test plan.

1.6 CLOSEOUT SUBMITTALS

A. Submit approved as-built data, final settings, test records, recommended spares, and operation and maintenance manuals.

PART 1 — GENERAL

1.7 QUALITY ASSURANCE

A. Manufacturer shall maintain an ISO 9001-certified quality system and demonstrate comparable data-centre experience.

B. Test instruments shall have current calibration traceable to a recognised laboratory.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Protect equipment from moisture, shock, corrosion, and contamination.

B. Maintain equipment-tag, purchase-order, serial-number, and source-lot traceability.

1.9 WARRANTY

A. Provide the project warranty commencing at Taking Over or the date stated in the Contract, whichever governs.

PART 2 — PRODUCTS

2.1 MANUFACTURERS

A. Subject to compliance with the Contract Documents, provide the system by an approved manufacturer.

B. Prime power rating shall be not less than **2500 kVA at 0.8 pf**.

2.2 PERFORMANCE REQUIREMENTS

A. Equipment shall operate continuously at the scheduled duty and site conditions.

B. Selections shall include stated tolerances, fouling allowances, harmonics, diversity, and redundancy as applicable.

C. Published catalogue values shall not replace project-condition certified data.

PART 2 — PRODUCTS

2.3 EQUIPMENT

A. Provide a complete factory-engineered assembly with coordinated ratings and interfaces.

1. Diesel engine and cooling system
2. Alternator, AVR and generator controller
3. Base frame, fuel system and synchronising controls

B. Materials shall be suitable for the operating environment and expected service life.

C. Frequency regulation shall not exceed **0.5 %**.

2.4 COMPONENTS

A. Components shall be accessible for inspection, isolation, removal, and replacement.

B. Safety-critical devices shall be independently identified and fail to a safe state.

C. Internal wiring, terminals, piping, and accessories shall be permanently identified.

D. Fuel autonomy at 100% load shall be not less than **24 h**.

PART 2 — PRODUCTS

2.5 CONTROLS AND INTERFACES

A. Hardwired safety and shutdown functions shall not depend solely on the supervisory network.

B. Provide scheduled BACnet/IP or Modbus TCP status, alarm, command, measurement, and setpoint objects.

C. Synchronise event timestamps to the project NTP service in Asia/Kolkata.

2.6 SOURCE QUALITY CONTROL

A. Perform documented incoming inspection, in-process checks, and final inspection.

B. Record material traceability, calibrated instrument numbers, torque values, and as-left settings.

PART 2 — PRODUCTS

2.7 FACTORY TESTING

A. Submit the factory test procedure before testing.

B. Test the offered rating and construction at the specified operating condition.

C. Record raw measurements, calculated results, instrument details, and acceptance status.

D. Do not claim equivalence for a different rating or condition without an approved engineering comparison.

PART 3 — EXECUTION

3.1 EXAMINATION

A. Verify foundations, access routes, clearances, environmental conditions, and embedded services before installation.

B. Do not proceed until unsatisfactory conditions are corrected.

3.2 INSTALLATION

A. Install in accordance with approved drawings and manufacturer instructions.

B. Install level, aligned, restrained, labelled, and accessible for operation and maintenance.

C. Coordinate earthing, power, controls, penetrations, fire stopping, drainage, and builder’s work.

PART 3 — EXECUTION

3.3 FIELD QUALITY CONTROL

A. Inspect the completed installation and record deficiencies.

B. Perform applicable continuity, insulation, pressure, leakage, rotation, interlock, and alarm tests.

C. Close major deficiencies before functional performance testing.

3.4 PERFORMANCE VERIFICATION

A. Demonstrate each specified performance criterion at stable scheduled conditions.

B. Load acceptance shall be not less than **80 % in one step**.

PART 3 — EXECUTION

3.5 COMMISSIONING

A. Execute L2 site acceptance, L3 pre-functional, L4 functional performance, and L5 integrated systems testing.

B. Coordinate test prerequisites, witnesses, temporary loads, trend logs, and restoration.

3.6 TRAINING AND CLOSEOUT

A. Train operations personnel using the installed equipment.

B. Submit signed test records, final settings, training attendance, and approved closeout documents.

END OF SECTION